

Joseph Weil

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Professional Summary

Detail-oriented professional with a B.S. in Applied Mathematics, along with experience in working with data analysis, statistical modeling, and automated trading systems. Experienced in analyzing large data sets, identifying performance trends, and developing data-based decision making frameworks. Seeking to leverage quantitative modeling and analytics experience to contribute to interdisciplinary work in energy resource planning and portfolio analysis.

Education

University of California, San Diego

Oct 2020 – Mar 2024

- B.S. in Mathematics – Applied Science
- Minor in German Studies
- 3.615 GPA

Relevant Professional Experience

Data Labeling Analyst II CW | Meta

June 2025 – Present

- Analyzed large datasets of annotation outputs to assess model robustness and policy compliance across iterations, contributing to LLM improvement and optimization through data-driven insights
- Identified policy vulnerabilities and model weaknesses through systematic risk evaluations and adversarial prompt probing, ensuring labeling accuracy and quality assurance across English and German datasets

Mathematics LLM Trainer | Outlier

Apr 2024 – May 2025

- Assessed AI-generated solutions to advanced undergraduate-level mathematical problems from subjects such as calculus and linear algebra, editing and rewriting solutions in LaTeX to ensure factual and notational conformity

Quantum ML Intern | Florence Quantum Labs Startup

Sep 2024 – Oct 2024

- Gained foundational knowledge in quantum computing concepts, including qubits, quantum gates, quantum algorithms, and machine learning concepts, with the goal of meteorological forecasting in the Qiskit framework

Projects - See personal website at the top

- **Autonomous Event Derivatives Trading System:** Developed an algorithmic prediction market trading bot in Python to pipeline market data, conduct quantitative analysis, forecast outcomes, and execute trades based on data-driven decisions.
- **iNaturalist API Observation Data Pipeline:** Developed a Python-based data pipeline to scrape and preprocess ecological observation data from the iNaturalist API, optimizing data handling for further analysis.
- **iNaturalist Data Case Study:** Conducted data cleaning, visualization, statistical analysis, and applied machine learning models on iNaturalist ecological data to both identify patterns in species distribution and biodiversity and ultimately assess ecological accuracy/data utility.
- **Personal Portfolio Website:** Designed and developed a personal website from scratch using HTML, CSS, and Javascript for the purposes of displaying personal projects.
- **Personal Budgeting Application:** Developing a personal budgeting application based in Python and the Pandas library for expense tracking, detailed financial analysis, and informed decision-making [in progress].

Skills and Languages

- **Technical:** Python, SQL, Pandas, Java, MATLAB, Microsoft Suite
- **Analytical:** Quantitative modeling, statistical analysis, numerical optimization, data engineering
- **Language:** German – Intermediate orally and fluent reading/writing
- **Professional:** Attention to detail, strong verbal and written skills, initiative to work independently, three years' experience in customer service

Additional & Coursework

Certification: Scientific Computing in Python – FreeCodeCamp

Oct 2024 – Nov 2024

Quantitative Coursework: Differential, Integral, Multivariate, and Vector Calculus for Engineering, Differential Equations, Real Analysis, Statistical Methods, Numerical Analysis: Linear algebra, Approximations, and Nonlinear Equations, Computer Science - Java, Macroeconomics series

Physics Coursework: Electricity and Magnetism, Climate and Atmospheric Dynamics, Optics, Mechanics, Thermodynamics, Fluids, Waves, Relativity and Quantum, Stellar Astrophysics, Black Holes, Ocean Waves

Extracurricular: UCSD Collegiate Sailing Team

Sept 2021 – Mar 2024